

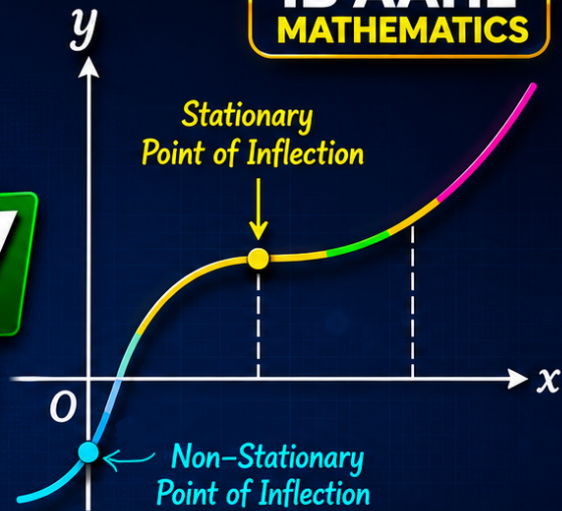
HOW TO FIND STATIONARY & NON-STATIONARY POINTS OF INFLECTION IN 4 EASY STEPS



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FOR **IB AAHL**

**IB AAHL
MATHEMATICS**



- ✓ Find y' and y''
- ✓ Find Critical Points
- ✓ Check Change in y''
- ✓ Identify Points of Inflection



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SL 5.8—Testing for max and min, optimisation. Points of inflexion



Question: Find stationary and non-stationary points of inflection of the following function:

$$f(x) = \frac{1}{4}x^4 - x^3$$

Step (1) : Differentiate the function:

$$f'(x) = \frac{1}{4}(4x^3) - 3x^2$$

$$f'(x) = x^3 - 3x^2$$

and $f''(x) = 3x^2 - 6x$

Step (2): Set $f''(x) = 0$ and find values of x .

$$f''(x) = 3x^2 - 6x$$

$$3x^2 - 6x = 0$$

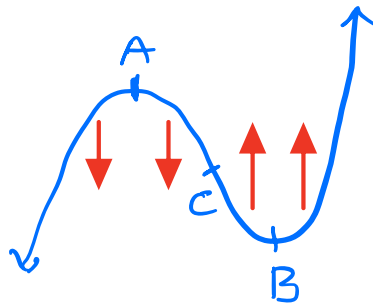
$$3x(x - 2) = 0$$

$$x = 0, x = 2$$

Note: How to identify maxima, minima and point of inflection?

If concavity remains same to the left and right of a point, then it is either max or min point.

And if concavity changes to the left and right of a point, then it is point of inflection.





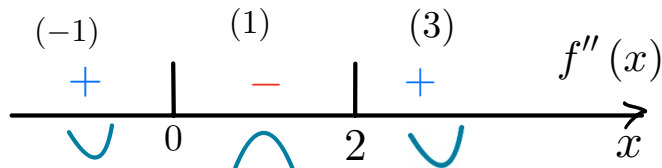
Step (3): Test for sign change(Sign diagram).

$$f''(x) = 3x(x - 2)$$

$$f''(-1) = 3(-1)(-1 - 2) = 9(+)$$

$$f''(1) = 3(1)(1 - 2) = -3(-)$$

$$f''(3) = 3(3)(3 - 2) = 9(+)$$



Sign of $f''(x)$ changes from (positive) + to (negative)- around $x = 0$

Hence $x = 0$ is point of inflection

And sign of $f''(x)$ changes from (negative) - to (positive)+ around $x = 2$

Hence $x = 2$ is point of inflection

Step (4): Classify using $f'(x)$:

$$\begin{aligned} f'(x) &= x^3 - 3x^2 \\ &= x^2(x - 3) \end{aligned}$$

at $x = 0$

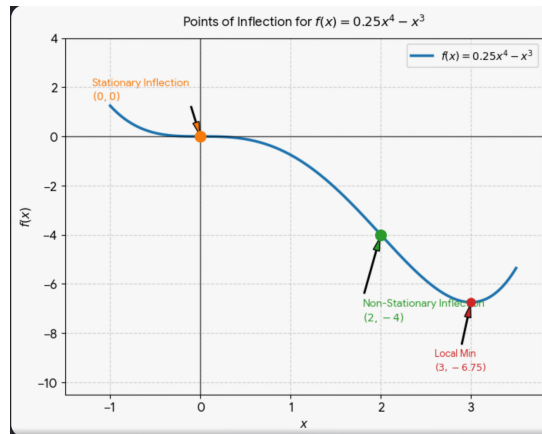
$$f'(0) = 0^2(0 - 3) = 0$$

since gradient is zero. Therefore, $(0, 0)$ is stationary point of inflection.

And

$$f'(2) = (2)^2(2 - 3) = -4 \neq 0$$

As gradient is not zero. Therefore, $(2, -4)$ is non-stationary point of inflection.



$$\left\{ \begin{aligned} f(x) &= \frac{1}{4}x^4 - x^3 \\ f(0) &= \frac{1}{4}(0)^4 - (0)^3 \\ f(2) &= \frac{1}{4}(-2)^4 - (2)^3 = -4 \end{aligned} \right.$$



Practice quiz: Let's test your skills with multiple choice practice quiz on finding and classifying stationary and non-stationary points of inflection.

1. Consider the function $f(x) = x^4 - 4x^3 + 10$. Find the x -coordinates of all points of inflection of $f(x)$.

A. $x = 0$ and $x = 3$

B. $x = 2$ and $x = 3$

C. $x = 0$ and $x = 2$

D. $x = 3$ only

2. Consider the function $f(x) = x^3$. What can be said about the point where $x = 0$?

A. It is a local maximum.

B. It is a local minimum.

C. It is a stationary point of inflection.

D. It is a non-stationary point of inflection.



3. Determine the nature of the point where $f''(x) = 0$ for the function $f(x) = x + \sin(x)$ on the interval $(0, 2\pi)$.

- A. Non-stationary point of inflection
- B. Local maximum
- C. Local minimum
- D. Stationary point of inflection

4. Consider the function $f(x) = x^3 + x$. What type of point occurs at $x = 0$?

- A. Stationary point of inflection
- B. Non-stationary point of inflection
- C. Local minimum
- D. Local maximum

5. Let $f(x)$ be a continuous function. If $f'(c) = 0$ and $f''(c) > 0$, what can be concluded about the graph of $f(x)$ at $x = c$?

- A. It has a local minimum.
- B. It has a local maximum.
- C. It is a point of inflection.
- D. It is a stationary point of inflection.



End of Lesson

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